Experiments

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Randomization Solves the Selection Problem

Recall the selection bias formula for diff. in means ($\tilde{\tau}$)

$$\begin{aligned}
\tilde{\tau} &= \mathbb{E}[Y_i|D_i=1] - \mathbb{E}[Y_i|D_i=0] \quad \text{(obs. diff. in means)} \\
&= \mathbb{E}[Y_{1i}|D_i=1] - \mathbb{E}[Y_{0i}|D_i=0] \\
&= \underbrace{\mathbb{E}[Y_{1i} - Y_{0i}|D_i=1]}_{\tau_{ATT}} + \underbrace{\mathbb{E}[Y_{0i}|D_i=1] - \mathbb{E}[Y_{0i}|D_i=0]}_{\text{Bias}}
\end{aligned}$$

How can we eliminate the bias term?

Random assignment of D_i will make the treated and untreated units identical on average, such that

$$\mathbb{E}[Y_{0i} \mid D_i = 1] = \mathbb{E}[Y_{0i} \mid D_i = 0]$$

This implies Bias $= 0$.

Are Experiments Feasible in Social Science?

- Large increase in the use of experiments in the social sciences: laboratory, survey, and field experiments
- *Abbreviated* list of examples (from Green 2008):
  - Program evaluation: development programs, education programs, SAT prep classes, weight loss programs, diversity training, deliberative polls, advertising campaigns, website designs...
  - Public policy evaluation: teacher pay, student incentives, class size, speed traps, vouchers, alternative sentencing, job training, health insurance subsidies, tax compliance, public housing
  - Behavioral research: persuasion, mobilization, education, income, interpersonal influence, conscientious health behaviors, media exposure, deliberation, discrimination
  - Research on institutions: transparency, corruption, electoral systems, information

Identification vs. Estimation

Goal of causal inference: Learn about a counterfactual quantity of interest (QoI) using *observed* data.

Causal inference thus involves two inferential hurdles:

1. Identification: If you can observe data from an entire *population*, can you learn about your QoI?
2. Estimation: Given your finite amount of data on a *sample*, how well can you learn about your QoI?

Golden rule of inference: IDENTIFICATION PRECEDES ESTIMATION

Classical Randomized Experiment

Setup:

- Units: $i = 1, ..., N$
- Treatment: $D_i \in \{0, 1\}$, randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}
- Observed outcome: $Y_i = Y_{D_i i}$
- Number of treated/untreated units: $N_1 = \sum_{i=1}^{N} D_i$ and $N_0 = N - N_1$

Notes:

- For now, we assume we have data on the entire population ($N =$ sample size $=$ population size)
- Random assignment can take one of several forms:
  - Complete randomization: Exactly N_1 treated units
  - Simple (Bernoulli) randomization: Each unit independently assigned to treatment with probability p

Identification of Average Treatment Effect Under Experiments

Identification Assumption: (guaranteed by random assignment):

$$\{Y_{1i},\ Y_{0i}\} \perp\!\!\!\perp D_i$$

Quantity of Interest:

$$\tau_{ATE} \equiv \mathbb{E}[Y_{1i} - Y_{0i}] = \frac{1}{N}\sum_{i=1}^{N}(Y_{1i} - Y_{0i})$$

How does our ID assumption lead to identification of τ_{ATE}?

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] &= \mathbb{E}[D_i \cdot Y_{1i} + (1 - D_i) \cdot Y_{0i}|D_i = 1] \\ &= \mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}] \quad \text{(by random assignment)}\end{aligned}$$

Similarly, $\mathbb{E}[Y_i|D_i = 0] = \mathbb{E}[Y_{0i}]$
So it follows that

$$\tau_{ATE} = \mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] = \underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{observed difference in means}}$$

Identification in Randomized Experiments

With $(Y_1, Y_0) \perp\!\!\!\perp D$, we can identify effect at any quantile:

Identfication Result

Let $F_{Y_d}(y)$ be the cumulative distribution function (CDF) of Y_d, then

$$\begin{aligned} F_{Y_0}(y) &= \Pr(Y_0 \leq y) = \Pr(Y_0 \leq y | D = 0) \quad (\textit{by randomization}) \\ &= \Pr(Y \leq y | D = 0). \end{aligned}$$

Similarly,

$$F_{Y_1}(y) = \Pr(Y \leq y | D = 1).$$

So the effect of the treatment at any quantile $\theta \in [0, 1]$ is identified:

$$\alpha_\theta = Q_\theta(Y_1) - Q_\theta(Y_0) = Q_\theta(Y|D = 1) - Q_\theta(Y|D = 0)$$

SATE and PATE

- Often we focus on estimating the average causal effect in a particular sample: **S**ample **A**verage **T**reatment **E**ffect (SATE)
  - uncertainty arises only from hypothetical randomizations.
  - inferences are limited to the sample in our study.
- Compare to the **P**opulation **A**verage **T**reatment **E**ffect (PATE)
  - requires knowledge about the sampling process that got you from population to sample
  - need to account for two sources of variation:
    - variation from the sampling process
    - variation from treatment assignment (randomness of D_i)
- If your sampling was random from a given population, then $\mathbb{E}[SATE] = PATE$. For this reason, we're often sloppy in distinguishing between the two quantities.

Uncertainty (SE) Estimates for SATE and PATE

A natural first attack: pretending

- Imagine two separate pops, one from which you get treated, one for control.
- Compute $\overline{Y}_1$ and $\overline{Y}_0$ based on these separate samples.
- Chance variation in the two then unrelated: $cov(\overline{Y}_1, \overline{Y}_0) = 0$

Get variance of the difference in means:

$$\begin{aligned}\mathbb{V}(\overline{Y}_1 - \overline{Y}_0) &= \mathbb{V}(\overline{Y}_1) + \mathbb{V}(\overline{Y}_0) + 2cov(\overline{Y}_1, \overline{Y}_0) \\ &= \frac{\sigma^2_{Y1}}{N_1} + \frac{\sigma^2_{Y0}}{N_0}\end{aligned}$$

$$SE_{ATE} = \sqrt{\frac{\sigma^2_{Y1}}{N_1} + \frac{\sigma^2_{Y0}}{N_0}}$$

where σ^2_{Y1} and σ^2_{Y0} are the variance of the Y_{1i} and Y_{0i} in the population

This is what we typically use, and it is

- the correct SE for the PATE
- what you get from t-tests (unequal variance), or regression with robust SE
- conservative for the SATE (where the sample is fixed but D_i is random)

Standard Error for Sample ATE

That made sense (I hope), but we cheated: can't sample treated and control independently since one group is the complement of the other, making $cov(\overline{Y}_1, \overline{Y}_0) \neq 0$

Standard Error for Sample ATE

Given complete randomization of N units with N_1 assigned to treatment and $N_0 = N - N_1$ to control, the true standard error of the estimated sample ATE is given by

$$SE_{\widehat{ATE}} = \sqrt{\left(\frac{N-N_1}{N-1}\right)\frac{\sigma^2_{Y1}}{N_1} + \left(\frac{N-N_0}{N-1}\right)\frac{\sigma^2_{Y0}}{N_0} + \left(\frac{1}{N-1}\right)2Cov[Y_1, Y_0]}$$

with population variances and covariances σ^2_{Yd}, $Cov(Y_1, Y_0)$.

Is this identifiable?

Standard error decreases if:

- N grows
- $\sigma^2_{Y1}, \sigma^2_{Y0}$ decrease
- $Cov[Y_1, Y_0]$ decreases

Proof: $SE_{\widehat{ATE}} \leq \widehat{SE}_{\widehat{ATE}}$ using conservative estimator

Upper bound for standard error is when $Cor[Y_1, Y_0] = 1$:

$$Cor[Y_1, Y_0] = \frac{Cov[Y_1, Y_0]}{\sqrt{Var[Y_1]Var[Y_0]}} \leq 1 \iff Cov[Y_1, Y_0] \leq \sqrt{Var[Y_1]Var[Y_0]}$$

$$\begin{aligned}
SE_{\widehat{ATE}} &= \sqrt{\left(\frac{N-N_1}{N-1}\right)\frac{Var[Y_1]}{N_1} + \left(\frac{N-N_0}{N-1}\right)\frac{Var[Y_0]}{N_0} + \left(\frac{1}{N-1}\right)2Cov[Y_1, Y_0]} \\
&= \sqrt{\frac{1}{N-1}\left(\frac{N_0}{N_1}Var[Y_1] + \frac{N_1}{N_0}Var[Y_0] + 2Cov[Y_1, Y_0]\right)} \\
&\leq \sqrt{\frac{1}{N-1}\left(\frac{N_0}{N_1}Var[Y_1] + \frac{N_1}{N_0}Var[Y_0] + 2\sqrt{Var[Y_1]Var[Y_0]}\right)} \\
&\leq \sqrt{\frac{1}{N-1}\left(\frac{N_0}{N_1}Var[Y_1] + \frac{N_1}{N_0}Var[Y_0] + Var[Y_1] + Var[Y_0]\right)}
\end{aligned}$$

Last step follows from the following inequality

$$\begin{aligned}
(\sqrt{Var[Y_1]} - \sqrt{Var[Y_0]})^2 &\geq 0 \\
Var[Y_1] - 2\sqrt{Var[Y_1]Var[Y_0]} + Var[Y_0] &\geq 0 \iff Var[Y_1] + Var[Y_0] \geq 2\sqrt{Var[Y_1]Var[Y_0]}
\end{aligned}$$

Proof: $SE_{\widehat{ATE}} \leq \widehat{SE}_{\widehat{ATE}}$, continued.

$$
\begin{aligned}
SE_{\widehat{ATE}} &\leq \sqrt{\frac{1}{N-1}\left(\frac{N_0}{N_1}Var[Y_1] + \frac{N_1}{N_0}Var[Y_0] + Var[Y_1] + Var[Y_0]\right)} \\
&\leq \sqrt{\frac{N_0^2 Var[Y_1] + N_1^2 Var[Y_0] + N_1 N_0 (Var[Y_1] + Var[Y_0])}{(N-1)N_1 N_0}} \\
&\leq \sqrt{\frac{(N_0^2 + N_1 N_0) Var[Y_1] + (N_1^2 + N_1 N_0) Var[Y_0]}{(N-1)N_1 N_0}} \\
&\leq \sqrt{\frac{(N_0 + N_1) N_0 Var[Y_1]}{(N-1)N_1 N_0} + \frac{(N_1 + N_0) N_1 Var[Y_0]}{(N-1)N_1 N_0}} \\
&\leq \sqrt{\frac{N\, Var[Y_1]}{(N-1)N_1} + \frac{N\, Var[Y_0]}{(N-1)N_0}} \\
&\leq \sqrt{\frac{N}{N-1}\left(\frac{Var[Y_1]}{N_1} + \frac{Var[Y_0]}{(N_0)}\right)} \\
&\leq \sqrt{\frac{N}{N-1}\left(\frac{\widehat{Var[Y_1]}}{N_1} + \frac{\widehat{Var[Y_0]}}{(N_0)}\right)}
\end{aligned}
$$

So the estimator for the standard error is conservative.

Seeing the Standard Error for Sample ATE

Want to know how ATE would differ under J randomizations of treatment, same sample: $SE_{\widehat{\theta}} \equiv \sqrt{\frac{1}{J}\sum_1^J(\widehat{\theta}_j - \overline{\widehat{\theta}})^2}$

i	Y_{1i}	Y_{0i}	Y_i	D_i	$P(D_i = 1)$
1	3	0	3	1	2/4
2	1	1	1	1	2/4
3	2	0	0	0	2/4
4	2	1	1	0	2/4

ATE estimates given all possible random assignments with two treated units:

Treated Units:	1 & 2	1 & 3	1 & 4	2 & 3	2 & 4	3 & 4
$\widehat{ATE}$:	1.5	1.5	2	1	1.5	1.5

Average $\widehat{ATE}$ is 1.5 and true standard error is $SE_{\widehat{ATE}} =$

$$\sqrt{\frac{1}{6}[(1.5-1.5)^2 + (1.5-1.5)^2 + (2-1.5)^2 + (1-1.5)^2 + (1.5-1.5)^2 + (1.5-1.5)^2]} \approx .28$$

Standard Error for Sample ATE

You can find the "true" formula is correct by using the (partially unobservable) quantities in the prior table to get:

$$
\begin{aligned}
SE_{\widehat{ATE}} &= \sqrt{\left(\frac{N-N_1}{N-1}\right)\frac{Var[Y_{1i}]}{N_1} + \left(\frac{N-N_0}{N-1}\right)\frac{Var[Y_{0i}]}{N_0} + \left(\frac{1}{N-1}\right)2Cov[Y_{1i}, Y_{0i}]} \\
&= \sqrt{\left(\frac{4-2}{4-1}\right)\frac{.25}{2} + \left(\frac{4-2}{4-1}\right)\frac{.5}{2} + \left(\frac{1}{4-1}\right)2(-.25)} \\
&\approx .28
\end{aligned}
$$

But remember, either formula or the table only work with unobservables.

Standard Error for Sample ATE

How does this compare to what we'll actually compute, on average?

i	Y_{1i}	Y_{0i}	Y_i
1	3	0	3
2	1	1	1
3	2	0	0
4	2	1	1

$\widehat{SE}_{\widehat{ATE}}$ estimates given all possible assignments with two treated units:

Treated Units:	1 & 2	1 & 3	1 & 4	2 & 3	2 & 4	3 & 4
$\widehat{ATE}$:	1.5	1.5	2	1	1.5	1.5
$\widehat{SE}_{\widehat{ATE}}$:	1.11	.5	.71	.71	.5	.5

The average $\widehat{SE}_{\widehat{ATE}}$ is $\approx$ .67 compared to true $SE_{\widehat{ATE}} \approx$ .28. In fact every one is bigger.

Example: Effect of Training on Earnings

- Treatment Group:
  - $N_1 = 7,487$
  - Estimated Average Earnings $\bar{Y}_1$: \$16,199
  - Estimated Sample Standard deviation $\widehat{\sigma}_{Y|D_i=1}$: \$17,038
- Control Group :
  - $N_0 = 3,717$
  - Estimated Average Earnings $\bar{Y}_0$: \$15,040
  - Estimated Sample deviation $\widehat{\sigma}_{Y|D_i=0}$: \$16,180
- Estimated average effect of training:
  - $\widehat{\tau}_{ATE} = \bar{Y}_1 - \bar{Y}_0 = 16,199 - 15,040 = \$1,159$
- Estimated standard error for effect of training:

$$\widehat{SE}_{\widehat{ATE}} = \sqrt{\frac{\widehat{\sigma}^2_{Y|D_i=1}}{N_1} + \frac{\widehat{\sigma}^2_{Y|D_i=0}}{N_0}} = \sqrt{\frac{17,038^2}{7,487} + \frac{16,180^2}{3,717}} \approx \$330$$

- Is this consistent with a zero average treatment effect $\alpha_{ATE} = 0$?

Testing the Null Hypothesis of Zero Average Effect

Null hypothesis, H_0: $\tau_{ATE} = 0$, i.e. $\mathbb{E}[Y_1] = \mathbb{E}[Y_0]$.

We observe a difference in mean earnings of $\hat{\tau}_{ATE} = 1,159$

What is the probability of observing a difference this large if the true average effect of the training were zero (i.e. the null hypothesis were true)?

- you may (should) know the answer already by comparing estimate to SE and a critical value
- let's try a t-test to formalize it

Testing the Null Hypothesis of Zero Average Effect

Use a two-sample t (or z) test with unequal variances:

$$t = \frac{\widehat{\tau}}{\sqrt{\frac{\widehat{\sigma}^2_{Y_i|D_i=1}}{N_1} + \frac{\widehat{\sigma}^2_{Y_i|D_i=0}}{N_0}}} = \frac{\$1,159}{\sqrt{\frac{\$17,038^2}{7,487} + \frac{\$16,180^2}{3,717}}} \approx 3.5$$

- We know that for large enough N, $t_N \xrightarrow{d} \mathcal{N}(0,1)$, for which the probability of observing a value of $|t|$ larger than 1.96 is $< .05$
- So obtaining a value as high as $t = 3.5$ is $p < 0.05$. In fact, very unlikely under the null:

```
> signif(2*(1-pnorm(abs(3.5))),3)
[1] 0.000465
```

- We reject the null hypothesis H_0: $\tau_0 = 0$ against the alternative H_1: $\tau_0 \neq 0$ at asymptotic 5% significance level whenever $|t| > 1.96$.
- Inverting the test statistic we can construct a 95% confidence interval

$$\widehat{\tau}_{ATE} \pm 1.96 \cdot \widehat{SE}_{\widehat{ATE}}$$

- What assumptions did we need along the way?

Testing the Null Hypothesis of Zero Average Effect

```
> d <- read.dta("jtpa.dta")
> t.test(earnings~assignmt,data=d,var.equal=FALSE)

        Welch Two Sample t-test

data:  earnings by assignmt
t = -3.5084, df = 7765.599, p-value = 0.0004533
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1807.2427  -511.6239
sample estimates:
mean in group 0 mean in group 1
       15040.50        16199.94
```

Regression to Estimate ATE in Experiments

Estimator (Regression with Experiments)

The ATE can be expressed as a regression coefficient:

$$
\begin{aligned}
Y_i &= D_i\, Y_{1i} + (1 - D_i)\, Y_{0i} \\
&= Y_{0i} + (Y_{1i} - Y_{0i})\, D_i \\
&= \underbrace{\bar{Y}_0}_{\alpha} + \underbrace{(\bar{Y}_1 - \bar{Y}_0)}_{\tau_{Reg}} D_i + \underbrace{\{(Y_{i0} - \bar{Y}_0) + D_i \cdot [(Y_{i1} - \bar{Y}_1) - (Y_{i0} - \bar{Y}_0)]\}}_{\varepsilon} \\
&= \alpha + \tau_{Reg} D_i + \varepsilon_i
\end{aligned}
$$

- Does this assume constant treatment effects?
- Our SE estimator allows different variance for $D = 1$ and $D = 0$.
  - Implies heteroskedasticity
  - Use "HC2" heteroskedasticity-robust variance:

$$
\hat{\sigma}^2_{HC2} = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} = \tilde{\mathbb{V}}(\tilde{\tau})
$$

Regression to Estimate the Average Treatment Effect

```
> library(sandwich)
> library(lmtest)
>
> lout <- lm(earnings~assignmt,data=d)
> coeftest(lout,vcov = vcovHC(lout, type = "HC2"))

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept) 15040.50     265.38 56.6752 < 2.2e-16 ***
assignmt     1159.43     330.46  3.5085 0.0004524 ***
--
```

Testing in Small Samples: Fisher's Exact Test

- Test of differences in means with large N:

$$H_0 : \mathbb{E}[Y_1] = \mathbb{E}[Y_0], \quad H_1 : \mathbb{E}[Y_1] \neq \mathbb{E}[Y_0] \text{ (weak null)}$$

- Fisher's Exact Test:

$$H_0 : Y_{1i} = Y_{0i}, \quad H_1 : Y_1 \neq Y_0 \ \ \forall i, \qquad \text{(sharp null of no effect)}$$

Key idea: Under the sharp null, we "observe" all potential outcomes – can compute what ATE would be under alternate randomizations

...which allows you to see many effect estimates all under this null.

Testing in Small Samples: Fisher's Exact Test

Key idea: Under the sharp null, we "observe" all potential outcomes – can compute what ATE would be under alternate randomizations

Let Ω be the set of all possible ways to assign treatments.

Fisher's exact test procedure:

1. Calculate a statistic $\hat{\theta}_{true}$ (e.g. difference in means) from original treatment assignment data
2. Obtain the null distribution of the statistic by calculating the same statistic $\hat{\theta}(\omega)$ under the sharp null for every possible (or many) ω in Ω
3. Compare $\hat{\theta}_{true}$ to the null distribution of $\hat{\theta}(\omega)$'s to see how "extreme" it is

Testing in Small Samples: Fisher's Exact Test

i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2	ω_3	ω_4	ω_5
1	3	3	1	1	1	0	0	0
2	1	1	1	0	0	1	1	0
3	0	0	0	1	0	1	0	1
4	1	1	0	0	1	0	1	1
$\widehat{\tau}_{ATE}$			1.5					
$\hat{\tau}(\omega)$			1.5	0.5	1.5	-1.5	-.5	-1.5

So *under null* $\Pr(|\hat{\alpha}(\omega)| \geq |\widehat{\tau}_{ATE}|) = 4/6 \approx .67$.

Which assumptions are needed? None! Randomization as "reasoned basis for causal inference" (Fisher 1935)

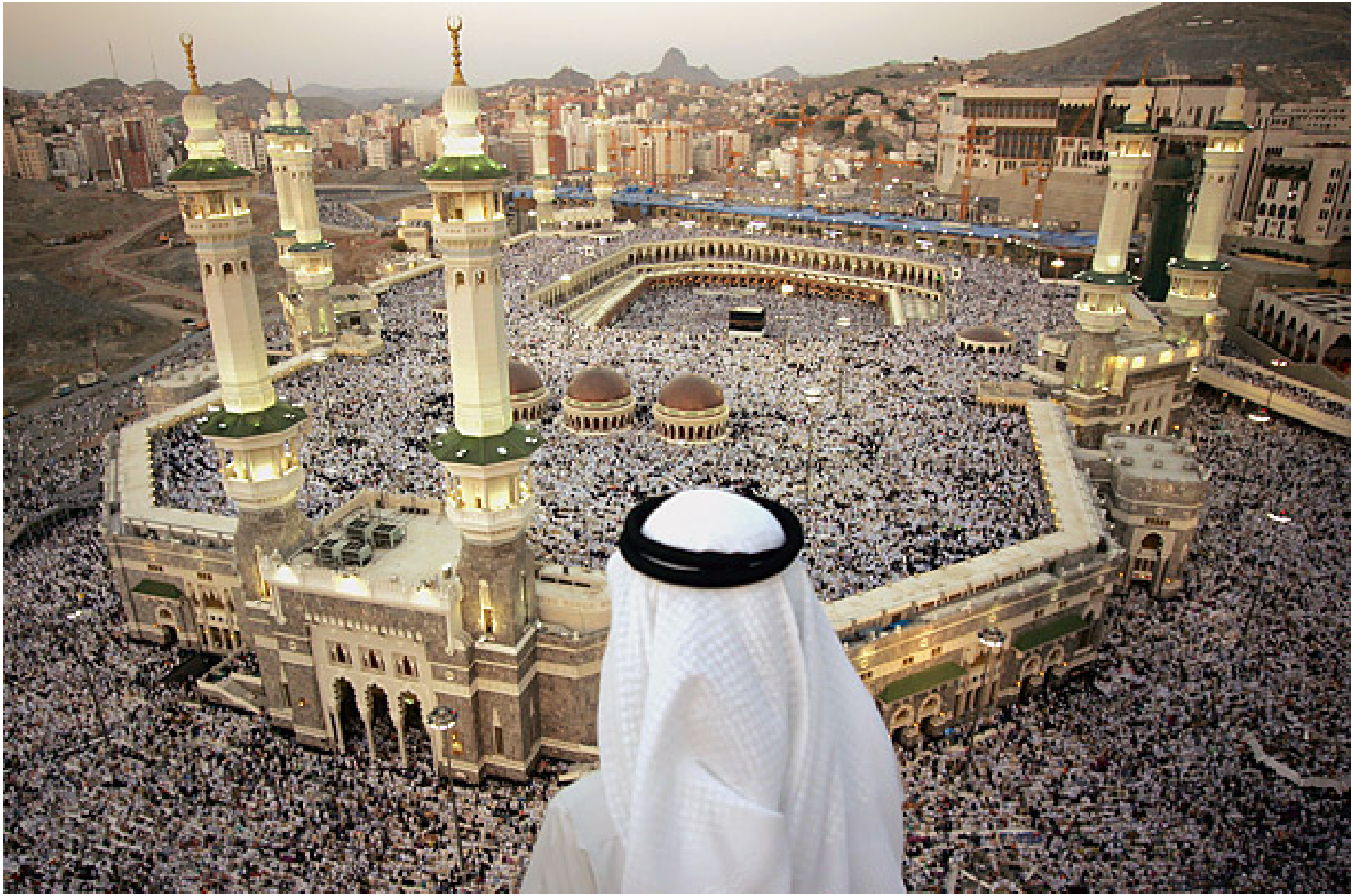
Threats to Internal and External Validity for Experiments

- Internal validity: can we estimate the treatment effect for our particular sample?
  - Fails when there are differences between treated and controls (other than the treatment itself) that affect the outcome
  - In other words, how might the randomization have failed? (See next)
- External validity: can we extrapolate our estimates to other populations?
  - Might Y_{1i}, Y_{0i}, and τ_i have looked different in a different part of the population?

Most Common Threats to Internal Validity

- Randomization fails at assignment
  - e.g. implementing partners assign their favorites to treatment group; imbalance due to small sample size
- Noncompliance with assignment
  - e.g. failure to treat or "crossover". Some members of the control group receive the treatment and some members of the treatment group go untreated
  - e.g. in JTPA: only 62% of women and 66% of men assigned for treatment actually enrolled in JTPA training (on the other side, compliance was almost perfect in the control group).
- Differential attrition
  - attrition *can* be okay, even if different rates for treated and contols
  - but, if probability of attrition depends on treatment status **and** potential outcomes, you break the randomization
  - Two scenarios: (i) people attrite from treatment-taking, but you still get to measure outcome; (ii) people attrite and you can't measure any outcome for them post-treatment.

Example: Clingingsmith, Khwaja, Kremer



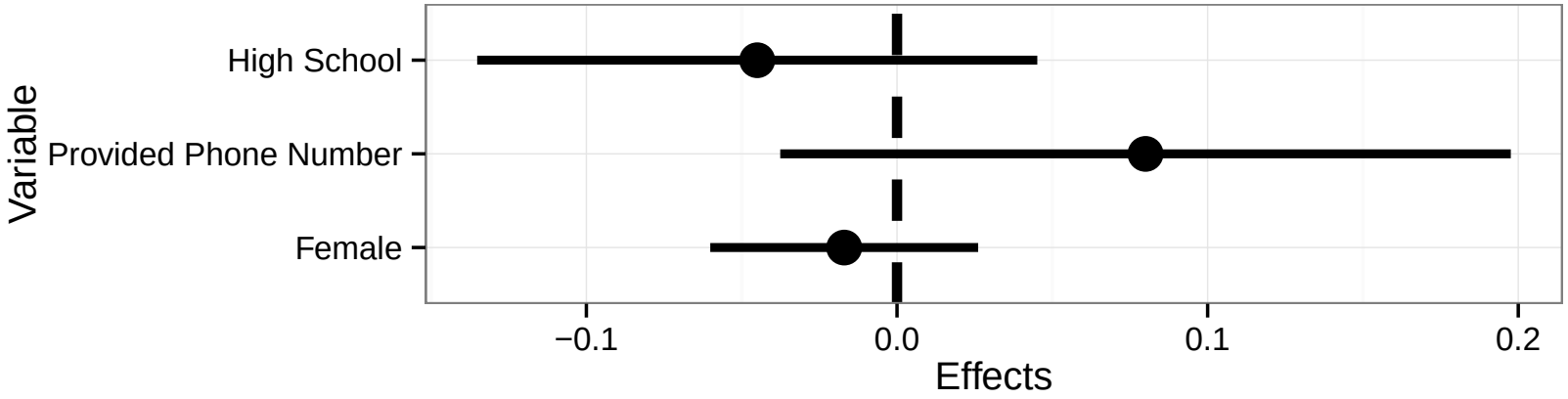
Example: Natural Experiment in Pakistan

- Pakistan allocated about 135,000 visas to Saudi Arabia for the Hajj via a randomized lottery.
- Wealthier Pakistanis tend to use private Hajj tour operators rather than the lottery.
- "Parties" apply to go, and are randomly selected, with stratification on sect, region, and accommodation.
- Compliance with the experiment is imperfect:
  - 99% who win lottery attend the Hajj.
  - 11% who lose lottery still attend the Hajj (via private tours).
- Since randomization not controlled by researcher, balance checks and qualitative checks are crucial!
- Remember, identifying assumption unprovable from the data alone so information on selection process is important.

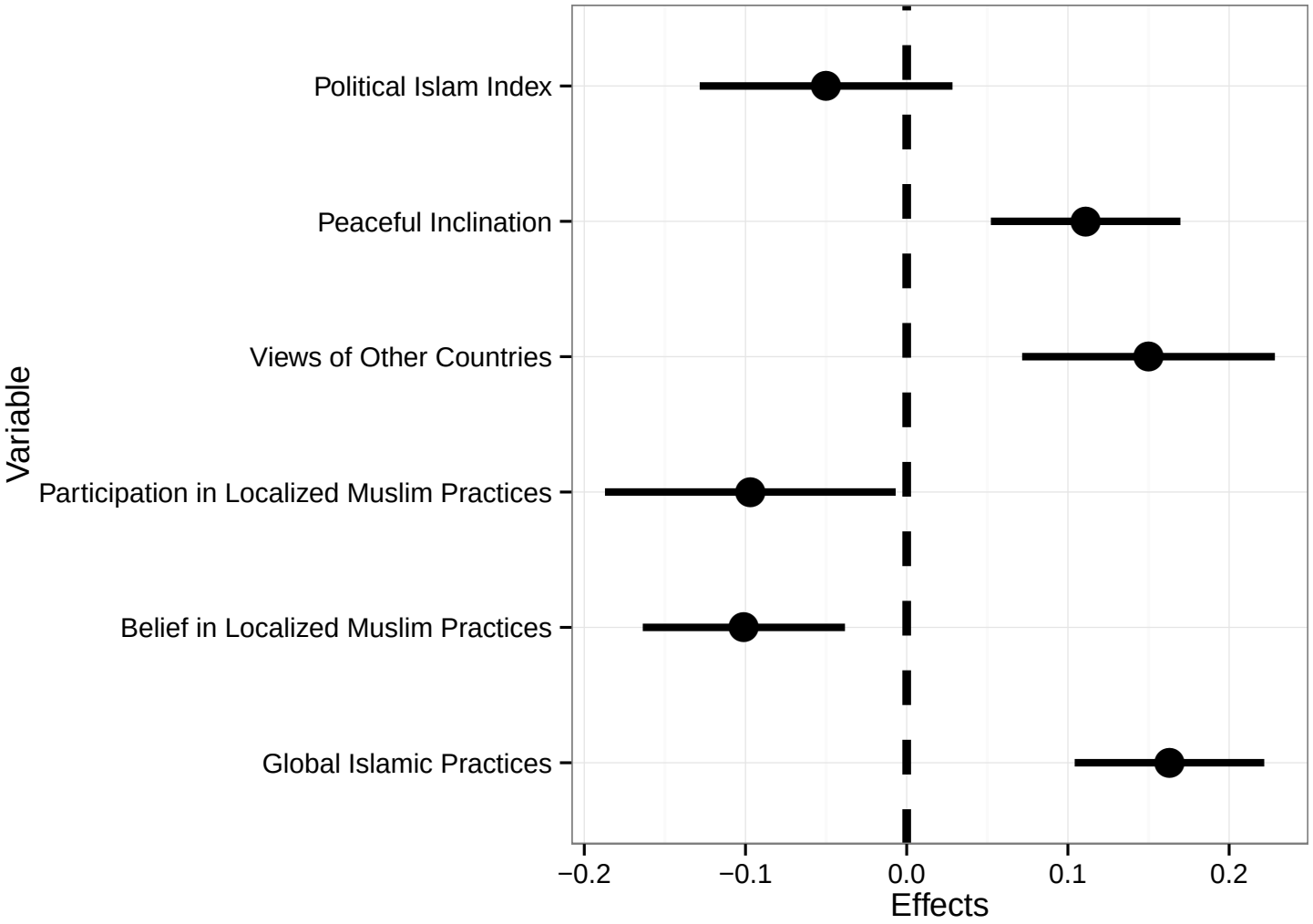
Example: Natural Experiment in Pakistan

Two kinds of information to bolster the randomization assumption:

1. Qualitative information:
   - The lottery selection algorithm was designed and implemented by an independent and reputable third party, and there were no reports of lottery manipulation.
2. Quantitative information: Balance tests on *observables*



Difference in Means: lottery winner vs. losers



Would you say these are the “effects of going on the Hajj”?

Most Common Threats to External Validity

- Non-representative sample
  - e.g. laboratory experiment using a convenience sample
  - units are randomly assigned, but not from the population of interest
- Non-representative treatment
  - the treatment differs from what you would actual implement
  - e.g. survey experiment about the effect of media priming on voting
  - scale effects
  - actual treatments may be bundled

Internal vs. External Validity

Which one is more important?

"One common view is that internal validity comes first. If you do not know the effects of the treatment on the units in your study, you are not well-positioned to infer the effects on units you did not study who live in circumstances you did not study." (Rosenbaum 2010, p. 56)

- Randomization (sometimes) ensures internal validity
- External validity may be partially addressed by comparing the results of several internally valid studies conducted in different circumstances and at different times
- While experiments can be especially narrow, same external validity issues often apply in observational work

That said, it is a balancing act

Often a studies with lower internal validity give us the theory, inspiration, qualitative knowledge needed for an experiment.

Concluding Remarks

- Random assignment solves the identification problem for causal inference based on minimal assumptions that the researcher can control
- Put differently, random assignment "balances observed and unobserved confounders", or "makes treated and control units comparable".
- Regression *can* be used for analyzing experiments: simple regression with robust SE yields unbiased estimates with conservative confidence intervals. Will be more useful (but tricky) when we add covariates.
- Possible tradeoff between internal validity and external validity
- What's Next: more on experimental design and analysis
  - covariate adjustment
  - block randomization